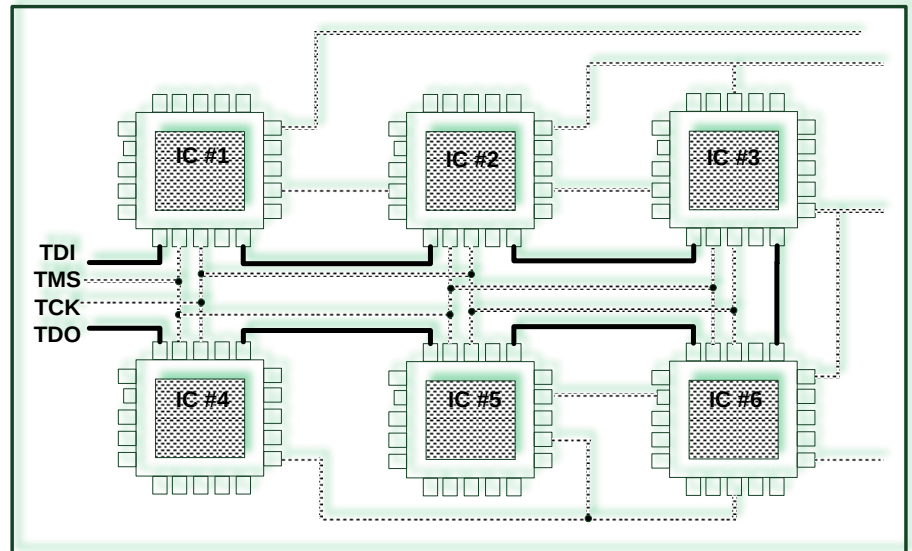


DFT – Part 2

- Introduction
- JTAG Architecture and Components
- JTAG Instructions
 - ◆ Mandatory Instructions
 - ◆ Optional Instructions
- Conclusion



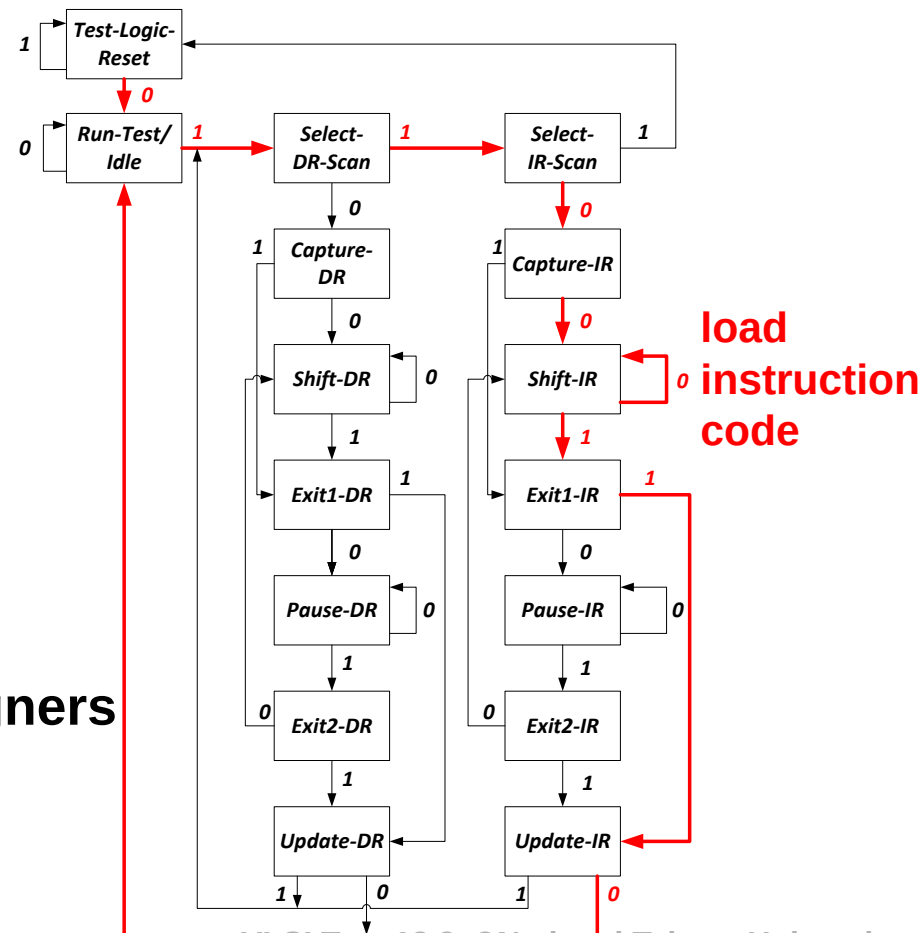
JTAG Instructions

- **Three Mandatory instructions:** must implement
- **Others are optional:** up to designer's decision

Instruction	M/O
EXTEST External test	Mandatory
BYPASS	Mandatory
SAMPLE / PRELOAD	Mandatory
INTTEST	Optional
RUNBIST	Optional
CLAMP	Optional
IDCODE	Optional
USERCODE	Optional
HIGHZ	Optional

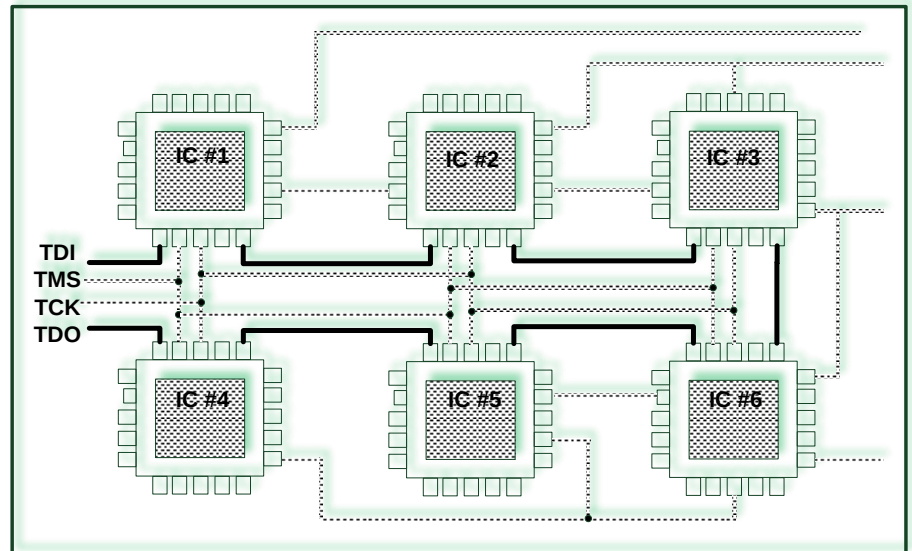
How to Load Instruction?

- Apply **TMS** sequence
 - ♦ **Initialize JTAG**: 1→1→1→ 1→1
 - ♦ **Load instruction**: 0→1→1→0→0→0.....0→1→1→0
- Number of zeros =
 - ♦ **Total length of Instruct. Reg.**
 - ♦ of all chips on board
- **Instruction code** is
 - ♦ shifted in via **TDI**
- JTAG specified Instruction codes
 - ♦ EXTEST=000...(all zeros)
 - ♦ BYPASS=111...(all ones)
 - ♦ Other codes specified by designers



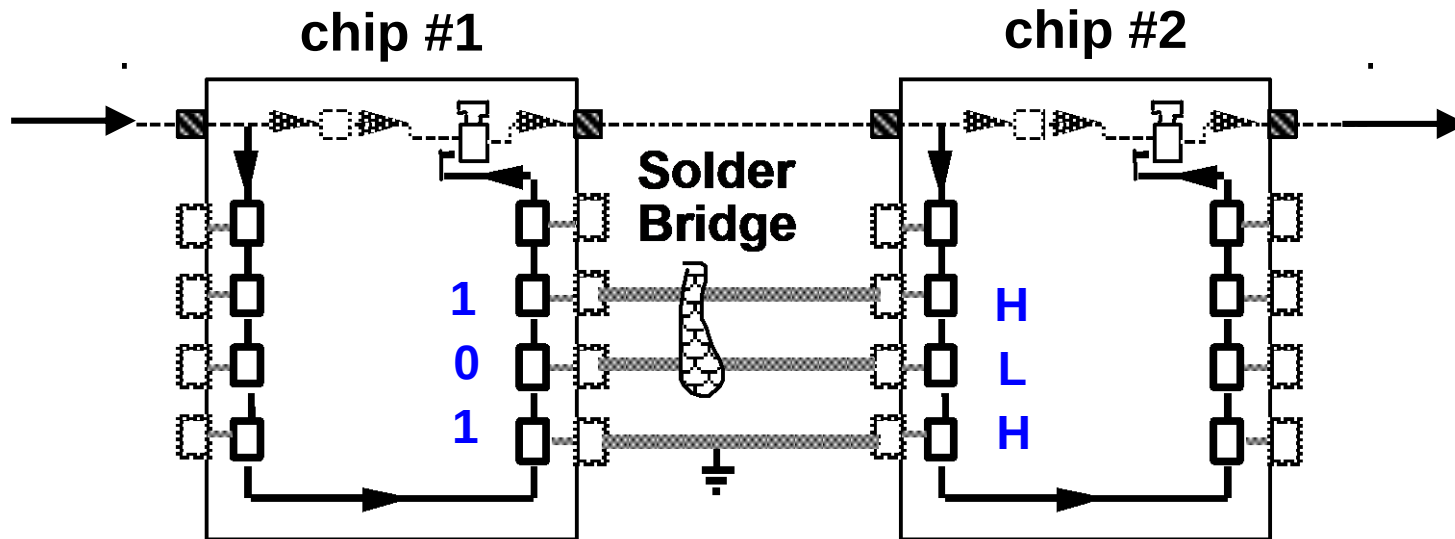
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EXTEST(1)

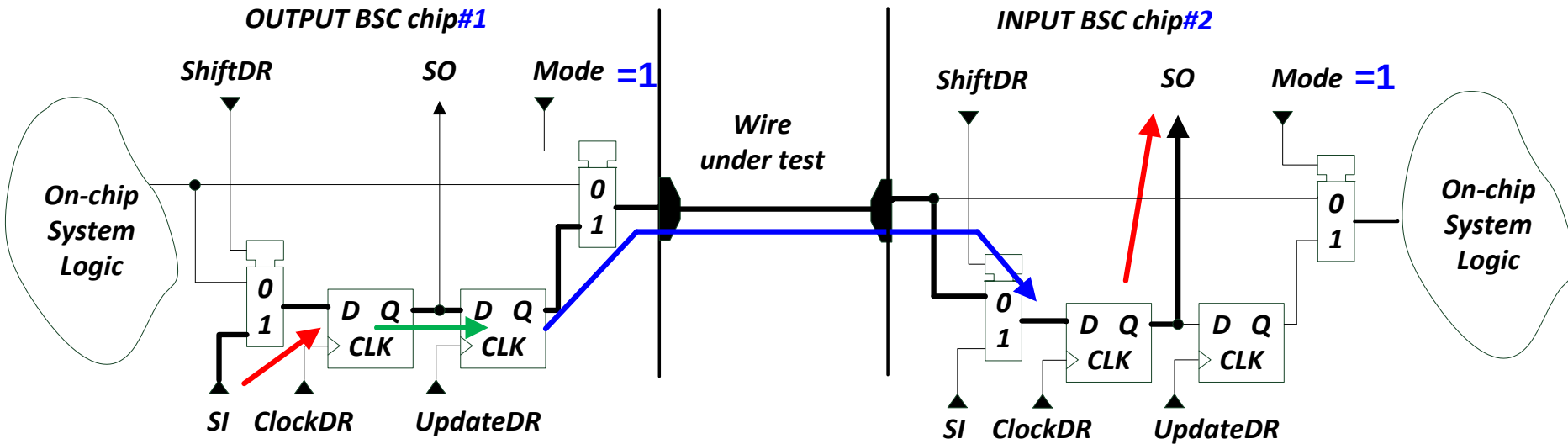
- Purpose: External test
 - ◆ Test **external** off-chip wire interconnections among chips



EXTEST Test Wires Among Chips

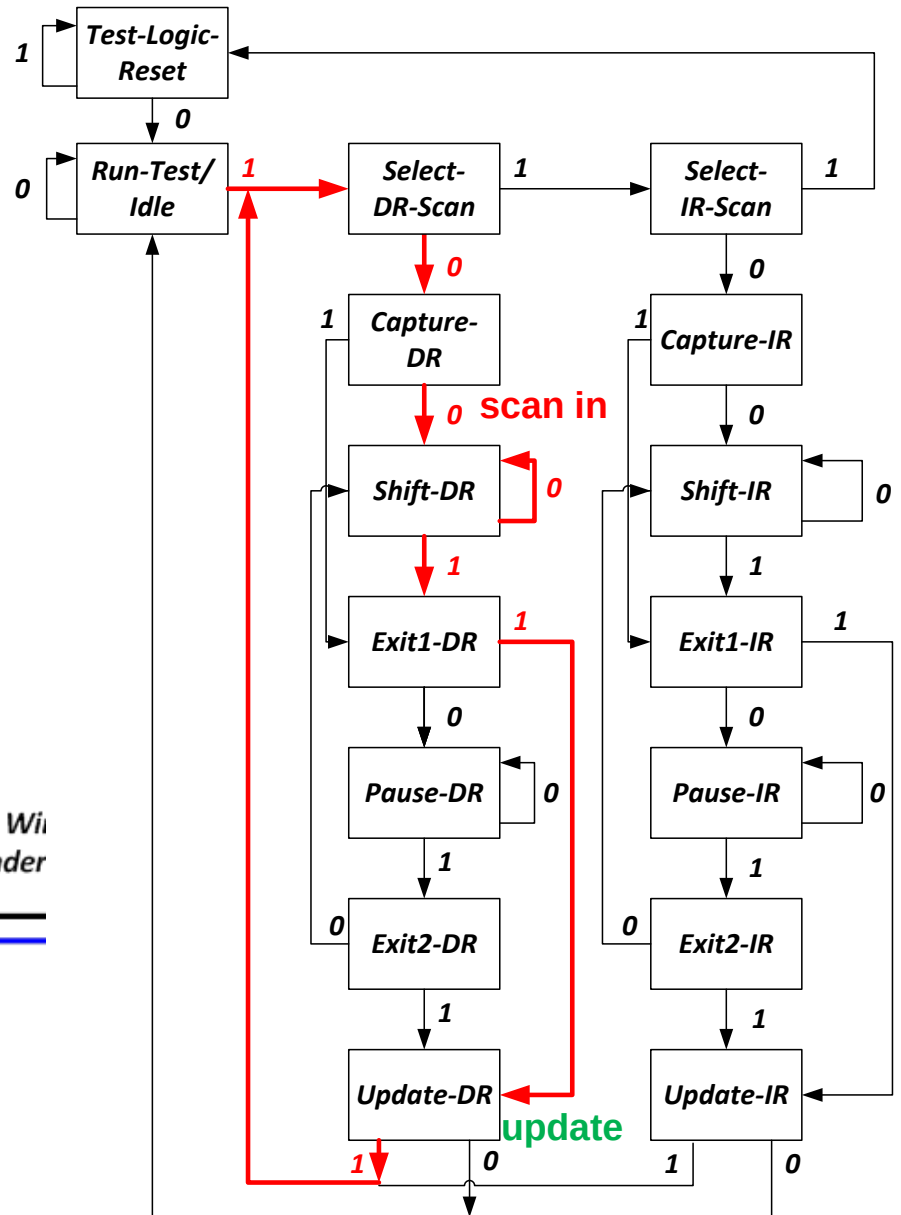
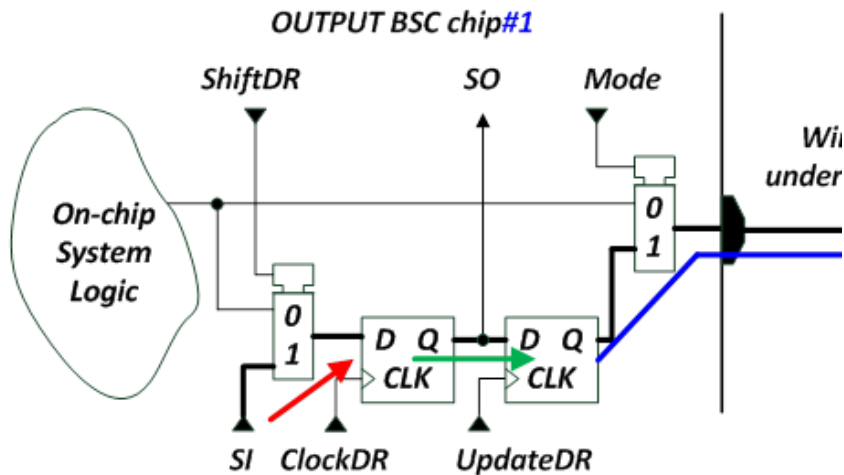
EXTEST (2)

- Step1: **Scan in chip#1**: shiftDR=1, clockDR...
- Step2: **Update chip#1**: updateDR
- Step3: **Capture chip#2**: shiftDR=0, clockDR
- Step4: **Scan out chip#2**: shiftDR=1, clockDR...
- Mode always = 1



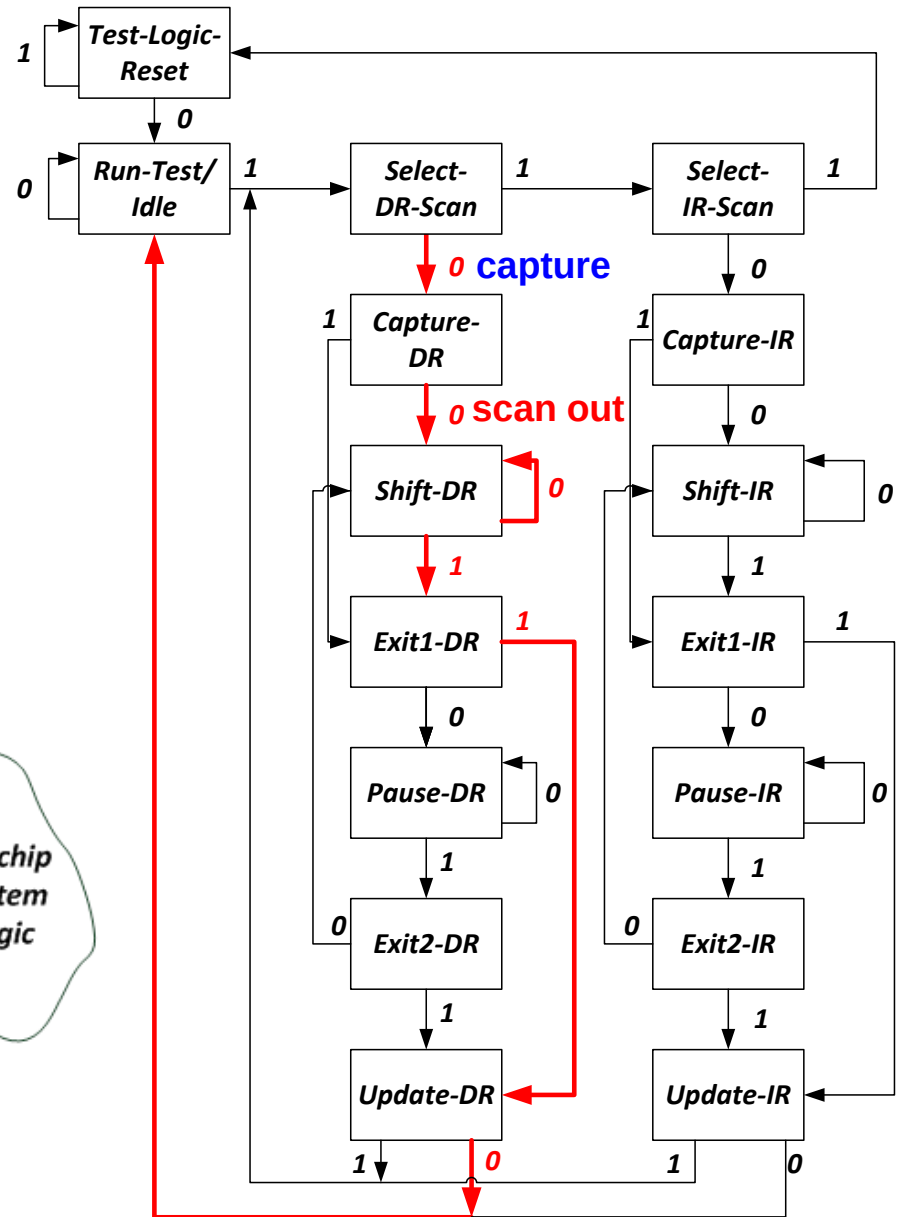
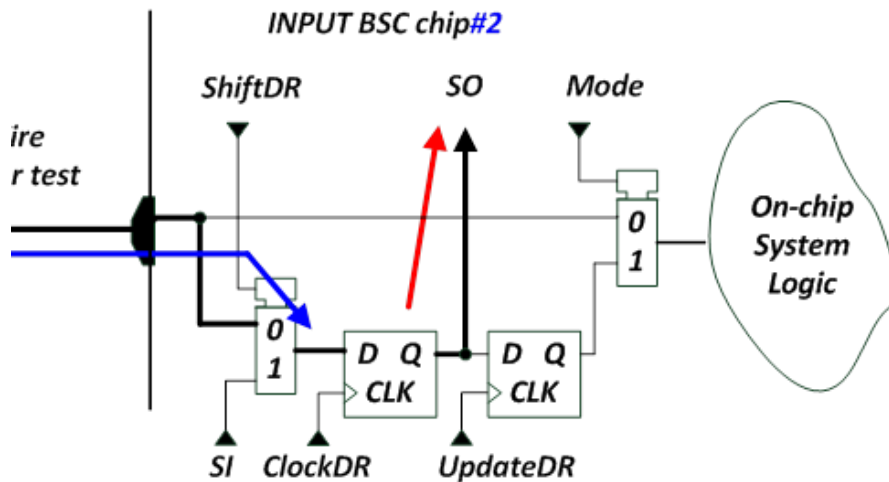
BSC and TAP Controller (1)

- (after instruction code loaded)
- TMS = 1→0→0→0...0→1→1→1
- Number of zeros =
 - ◆ Total length of Data Reg.
 - ◆ NOTE: BSR, BR



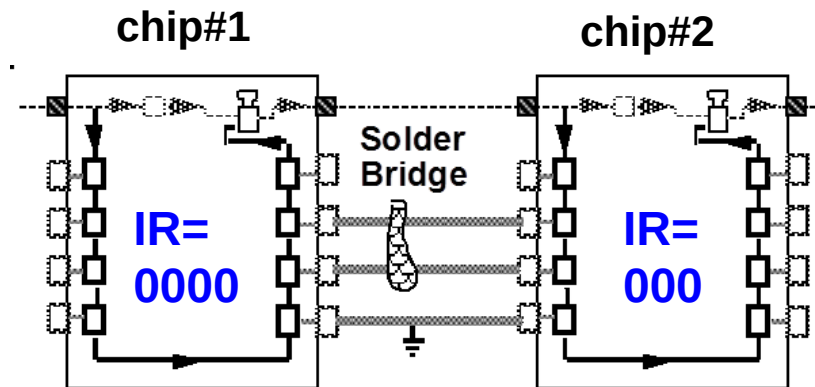
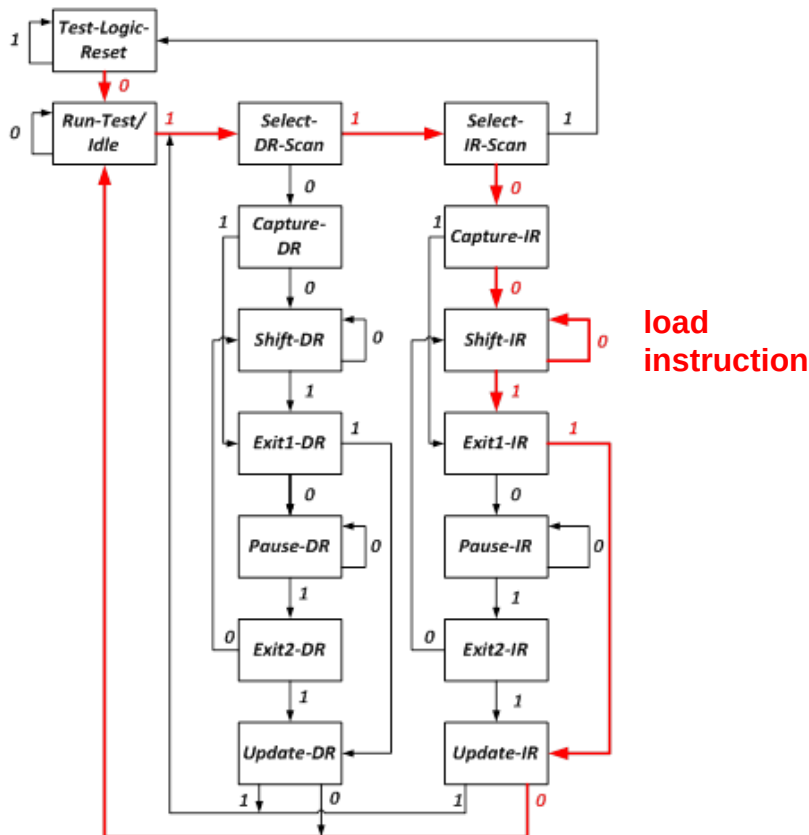
BSC and TAP Controller (2)

- TMS = 0 → 0 → 0 ... 0 → 1 → 1 → 0



Example (1/3) Load Instruction

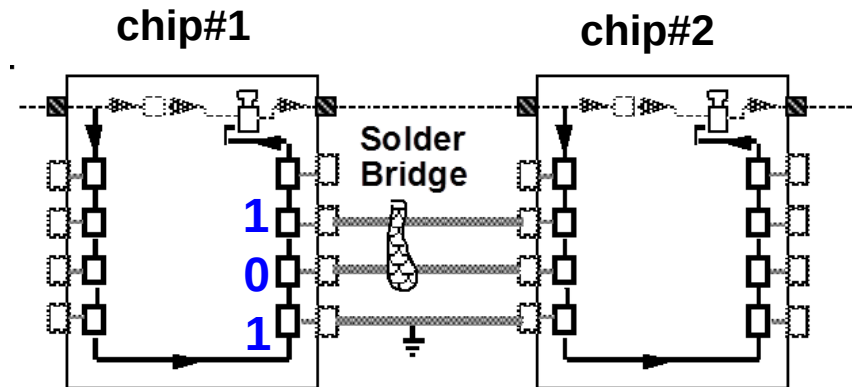
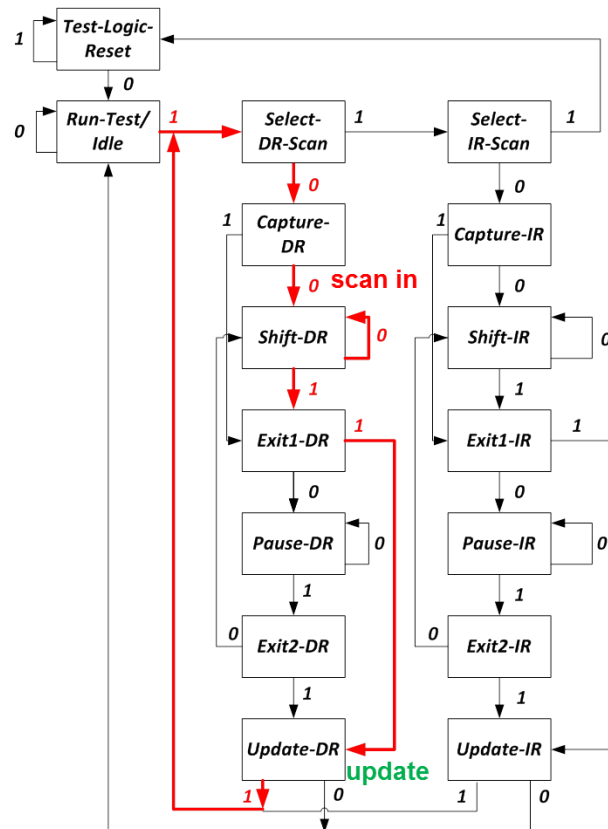
	Initialize TAP		load instruction	
TMS	11111	01100*	000 0000**	110
TDI			000 0000	
TDO				
Final state	test logic-reset		Shift-IR	Run-test/Idle



- * left bit shifted in first
- **chip#1 IR has 4 bits
chip#2 IR has 3 bits
total length of IR = 7

Example (2/3) Scan-In-Update

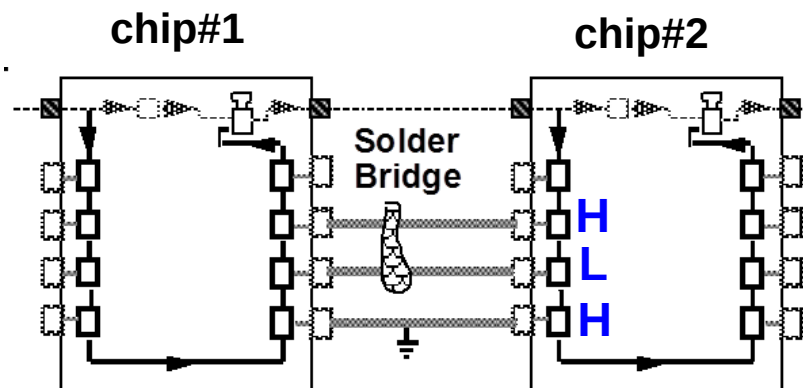
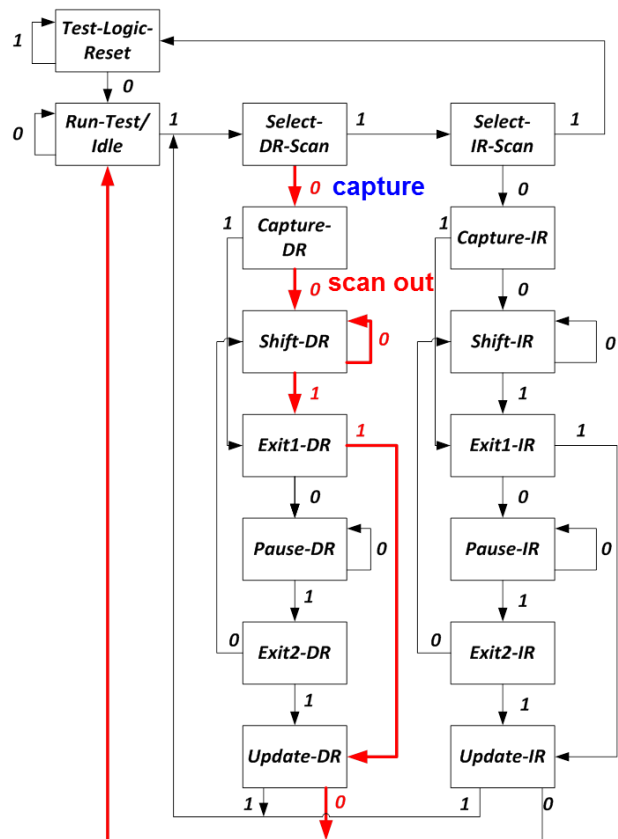
		Scan in		Update	
TMS	100	0000 0000 0000 0000	1	1	1
TDI		XXXX XXXX X 101 XXXX *			
TDO					
Final state		Shift-DR		update-DR	select-DR-scan



* left bit shifted in first
chip#1 DR has 8 bits
chip#2 DR has 8 bits
total length of DR = 16

Example (3/3) Capture-Scan-Out

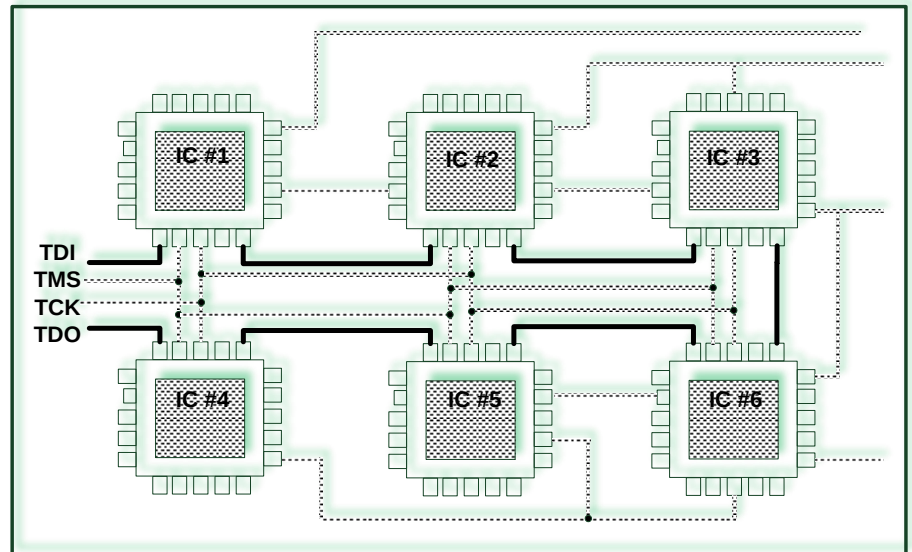
	Capture		Scan out	
TMS	0	0	0000 0000 0000 0000	110
TDI				
TDO			XXXX HLH X XXXX XXXX *	
Final state	Capture-DR		Shift-DR	run-test/Idle



* left bit shifted out first
 chip#1 DR has 8 bits
 chip#2 DR has 8 bits
 total length of DR = **16**

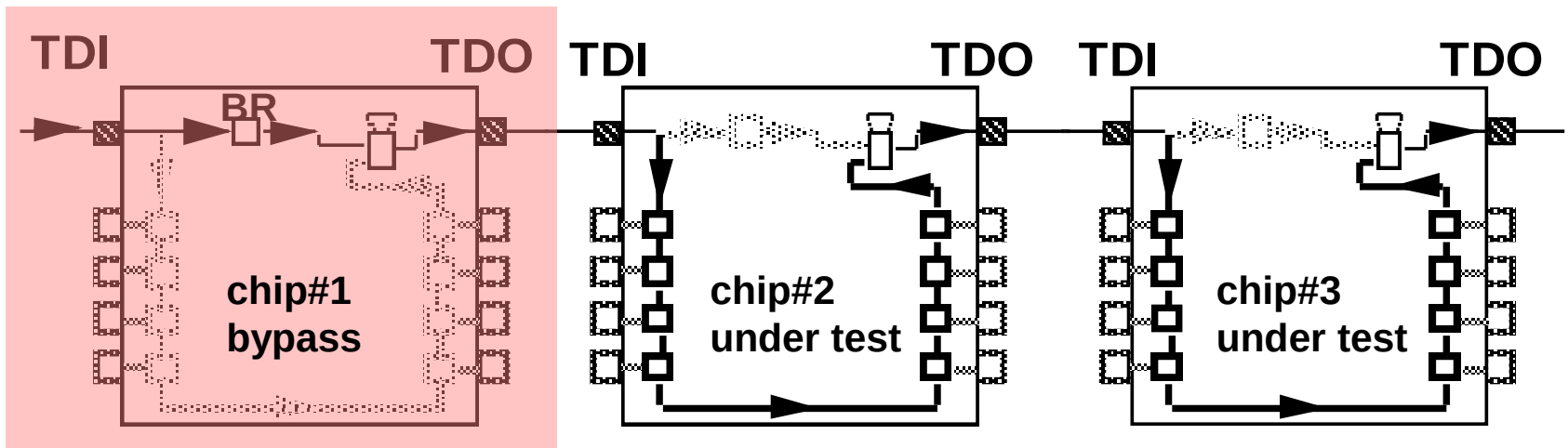
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Bypass Instruction

- Purpose: bypass scan data from TDI to TDO of a chip
 - ♦ Through **bypass register (BR, one bit)**
- Example:
 - ♦ Go through chip#1: 24 clocks
 - ♦ Bypass chip#1: $1+8+8=17$ clocks



Bypass Saves Test Time

Quiz

Q: Suppose we add chip#1 to our board. We want to apply EXTEST to chip#2 and chip#3. Please modify TMS/TDI/TDO sequence to bypass chip#1.

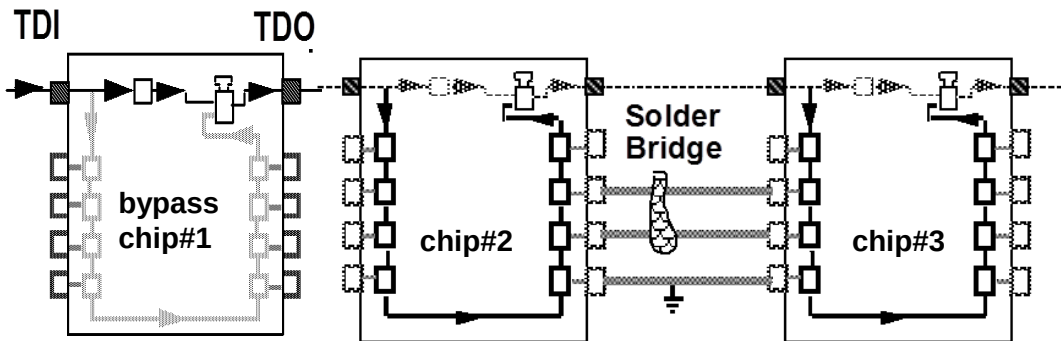
NOTE:

BYPASS instruction code for chip#1 = '11'
EXTEST instruction code '000' and '0000'

	Init. TAP		load instruction	
TMS	11111	01100	000 0000	110
TDI			000 0000	
TDO				
state	reset		Shift-IR	Idle

	Scan in	update
100	0000 0000 0000 0000	111
	XXXX XXXX x101 XXXX	
	Shift-DR	

Capture	Scan out	
00	0000 0000 0000 0000	110
	XXXX HLHx XXXX XXXX	
Capture-DR	Shift-DR	Idle



Summary

- Two mandatory instructions
 - ◆ **EXTEST** (000...0)
 - * Test wires among chips
 - * scan-in, update, capture, scan-out
 - ◆ **BYPASS** (111...1)
 - * Bypass chips to save test time

